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Plastic Behavior of 3-Bromo-5-chlorobenzoic Acid: Structural, Thermal, and Plastic Deformation Analysis

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ABSTRACT: Plastic crystals are a unique class of dynamic crystalline materials that combine long-range order with partial molecular mobility, resulting in remarkable mechanical properties, such as plasticity and flexibility. These crystals have emerged as promising candidates for advanced flexible devices and pharmaceutical applications. In this study, we investigate the plastic behavior of 3-bromo-5-chlorobenzoic acid (3B5CBA), a globular halogenated aromatic compound. Crystals of 3B5CBA grown via a fast evaporation technique exhibited plastic bending when stressed along the (001) plane. Thermal analysis of crystals revealed a sublimation onset at approximately 70 °C and a melting point of 195.3 °C. Single-crystal X-ray diffraction (SCXRD) analysis and energy framework calculation revealed strong 2D sheet-like interactions parallel to the (001) plane (−299.9 kJ/mol) and weaker interlayer C−Br···O interactions (−79.6 kJ/mol) forming low-energy slip planes. Nanoindentation on the (001) face confirmed homogeneous plastic deformation with a moderate hardness of 217.48 ± 19.87 MPa and a Young's modulus of 2.94 ± 0.35 GPa. Furthermore, shape analysis using Hirshfeld surface parameters showed high globularity ($G = 0.808$) and low asphericity ($\Omega = 0.108$). Together, these results showed that 3B5CBA being an organic aromatic compound with a nearly globular shape, hydrogen and halogen synthon-forming functional groups on its periphery can show long-range disordered crystal structure, sublimation ability, and irreversible plastic deformation, offering new insights into the design of aromatic plastic molecular solids.

1. INTRODUCTION

Plastic crystals (PCs) represent a distinct category of dynamic crystalline solids composed of molecules that retain the ability to rotate within the crystal lattice.[1,2] Known since the early 20th century, the plastic crystalline phase—also referred to as the orientationally disordered phase[2]—is characterized by partial molecular mobility despite the presence of long-range order. Traditionally, PCs are formed from nearly globular molecules that are either symmetrical about their center or can approximate a spherical shape through rotational motion.[2−6] However, recent studies have revealed that the mechanical properties of molecular crystals depend heavily on their molecular arrangement and intermolecular interactions,[7] which can also give rise to plastic crystals, provided their molecular geometries allow for minimal steric hindrance to rotation. PCs exhibit a set of unusual yet fascinating properties, including solid−solid phase transitions before melting,[2,5,6,8,9] rotational dynamics in the solid state,[1,2,10] orientational disorder,[2,11,12] and long-range structural order like traditional crystals.[1] Additional characteristics include sublimation behavior,[3] high compressibility, and remarkable mechanical plasticity.[1,2,6,9,12,13]

Developing the mechanical properties of organic crystals is a rapidly advancing field.[14−18] Due to their diverse mechanical behaviors, growing interest in flexible organic crystals has garnered attention from crystallographers and materials scientists. The mechanical properties of molecular crystals are typically categorized as reversible or irreversible, based on whether the crystal returns to its original shape after the application and release of applied mechanical force. Elastic bending is a reversible property,[19−27] as crystals return to their original shape once the applied force is removed, whereas plastic bending and brittleness are irreversible properties, leading to permanent deformation.[18,20,21,28,29] Elasticity, plasticity, and brittleness in crystals are influenced by their intrinsic properties and the rate at which the load is applied. These mechanical characteristics of organic crystals are the effect of molecular packing that is led by noncovalent interactions.[18,21,26,28−30] Some molecular crystals exhibit face-specific plasticity where deformation occurs only on particular crystal faces. Plastic bending, in turn, can be classified into one-dimensional (1D) plastic bending,[18,20,28,31] where deformation is restricted to a single direction, and two-dimensional (2D) plastic bending,[15,28] which encompasses more complex deformations such as twisting.[19,23,32,33] Flexible organic crystals

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hold significant potential in a range of applications, including mechanical actuators,[34] organic electronics,[35−37] optical devices,[38,39] and pharmaceuticals.[40−43] Notably, in the pharmaceutical industry, plastically bendable crystals have gained prominence due to their superior powder compaction and tablet-forming capabilities,[44] which are critical for drug formulation.[43,45,46]

The mechanical behavior of molecular crystals is largely influenced by their molecular packing and the nature of intermolecular interactions. Plasticity in organic crystals typically exhibits a combination of weak and strong intermolecular forces, such as van der Waals interactions,[19,21,47] hydrogen bonds,[24,30,32] halogen bonds,[24,28,30] and $\pi-\pi$ stacking.[33,46,48] Weak and strong intermolecular forces in an orthogonal direction create the anisotropic interaction in the crystal, responsible for plastic bending.[18,20,22,28,30] The weak intermolecular interactions contribute to the flexibility of the crystals by acting as buffers (between the molecular layers) that dissipate mechanical strain. In plastic bending, mechanical stress is accommodated through the sliding of molecular layers along slip planes, facilitated by weak intermolecular interactions;[18,21,46,49] the interaction energy along these planes is very low because of their weak interaction. These low-energy slip planes can be achieved through strategic crystal engineering.[14−16] Desiraju et al. for the first time demonstrated the plastic deformation of hexachlorobenzene and 2-methylthionicotinic acid, where the slip-plane movement was attributed to weak dispersive interactions.[50] Mondal et al. highlighted the importance of a high globularity and dispersive nature in promoting mechanically soft plastic crystal phase in three amino boranes: BH_3NMe_3, BF_3NMe_3, and BH_3NHMe_2. Despite comparable globularity and asphericity, BH_3NHMe_3 showed less plasticity (restricted to 1D plasticity), which was attributed to the presence of strong hydrogen bonding interactions that limit the number of available slip planes. While the other two solids, BF_3NMe_3 and BH_3NMe_3, were relatively more plastic and were crystallized in a high symmetry space group ($R3m$) possessing a large number of slip planes. As a result, these crystals exhibited exceptional ductility and demonstrated 3D plastic behavior reminiscent of conventional metals like gold.[12] Recently, Dutta et al. proved that globular bicyclic phosphate oxides and sulfides featuring methyl and ethyl tails can possess plastic and elastic deformation. Head-to-tail arrangement of molecules formed molecular columns, which were responsible for multiple slip planes, leading to plastic deformation, and corrugated sheets formed by weak interconnected columns were responsible for forming an isotropic internal structure, which is responsible for elastic deformation.[51]

In this work, we report observations related to the plastic behavior of 3-bromo-5-chlorobenzoic acid (3B5CBA) crystals. While plastic[14] and elastic[24] behavior of halogenated benzoic acids, namely, 2,3-dichlorobenzoic acid, 3,4-dichlorobenzoic acid, 2-chlorobenzoic acid, and 2-bromobenzoic acid, have been reported, our work is the first report on the mechanical behavior of 3B5CBA crystals. Further, none of the previous studies on halogenated benzoic acid have reported quantitative mechanical properties such as the elastic modulus (E) and hardness (H) of halogenated benzoic acid crystals. Our work is the first such report to provide measurements for any halogenated benzoic acid crystals using nanoindentation. In this work, various solid-state material characterization techniques have been used to analyze the structural, thermal, and mechanical behavior of 3B5CBA. Energy framework calculations have also been performed to provide insights into the anisotropic interaction energies within the crystal, enabling identification of potential slip planes governed by weak intermolecular forces.

2. MATERIALS AND METHODS

2.1. Materials Used. 3-Bromo-5-chlorobenzoic acid (3B5CBA) (98% pure) was procured from BLD Pharmatech (India) Pvt Ltd. Dichloromethane and chloroform of extrapure grade were purchased from Finar Chemical Ltd. Toluene was purchased from Aceto Pharma (India) Pvt. Ltd. Acetic acid, diethyl ether, acetone, acetonitrile, and methanol were purchased from Merck Life Science Pvt. Ltd. Ethyl acetate was purchased from SRL Chemical Ltd. Ethanol was purchased from Changshu Hongsheng Fine Chemical Co. Ltd. Benzene was purchased from RFCL Ltd. All solvents were used for crystallization without any additional purification.

2.2. Single-Crystal Formation and Characterization. Single crystals of 3-bromo-5-chlorobenzoic acid (3B5CBA) were grown via a fast evaporation technique using various solvents such as dichloromethane, acetone, diethyl ether, ethyl acetate, methanol, ethanol, acetic acid, chloroform, acetonitrile, toluene, and benzene (Table S1). Among these, long, high-quality needle-shaped crystals suitable for plastic deformation studies were obtained from the DCM solution of 3B5CBA (concentration of 5.5−6.0 mg/mL) after 2−3 days, when the solution was kept for solvent evaporation at room temperature. Cross-polarized light microscopy, optical microscopy, and hot-stage microscopy were employed to select pristine needle-shaped single crystals, capture videos of plastic deformation, and examine the thermal response of 3B5CBA crystals, respectively. These analyses were carried out using an optical microscope (Nikon Eclipse LV100 NPOL), a LINKAM thermal stage (model: THMS600, 11002−2186), and a Photron high-speed camera (Fastcam MINI UX 100) with a 10 μm pixel pitch.

The surface morphology of both bent and unbent segments of 3B5CBA single crystals was examined by using field emission scanning electron microscopy (FESEM, JSM 7600F, JEOL, Japan). Imaging was conducted at various magnifications to assess surface defects across different segments of the crystals. In addition, morphology-based characterization of the bent and unbent crystals was performed by using AFM. For AFM analysis, crystals were affixed onto clean glass slides using the drop-casting method and subsequently dried in a desiccator. Imaging was conducted in tapping mode using AFM (Bruker), and the acquired data were processed using JPK software. PXRD diffraction of samples was recorded on D8 Discover, Bruker AXS, GmbH, in the range of 5−50°. Raman spectra of samples were recorded on A KAISER RXN1 with an acquisition time of 5 s for reference. A Raman spectrometer uses a laser with an excitation wavelength of 785 nm and a maximum power of 400 mW. FTIR spectra of samples were recorded using an INVENIO S spectrometer (Bruker) over the range of 4000−400 cm^{-1}. Simultaneous DSC and TGA were conducted on a NETZSCH STA 449 F3 Jupiter, Germany. About 2−3 mg of sample was placed in perforated aluminum crucibles for analysis with a 5 K/min heating rate in the temperature range of 30−250 °C under a dry nitrogen atmosphere (at 40 mL/min). NETZSCH was used to determine the onset and peak temperatures of the curves.

2.3. Single-Crystal X-ray Diffraction and Characterization. A pristine single crystal was selected for structural analysis using an optical microscope and mounted on the goniometer of a Bruker D8 QUEST diffractometer equipped with a Mo Kα radiation source (λ = 0.71073 Å). Data collection was carried out at 298.15 K, and absorption corrections were applied using the multiscan method (SADABS). The crystal structure was solved using the SHELXT[52] program through intrinsic phasing and refined using the SHELXL refinement package with least-squares minimization, all implemented within the Olex2 software suite.[53] All non-hydrogen atoms were refined anisotropically. Halogen atoms in the asymmetric unit were disordered over two positions with site occupancy. The disorder was

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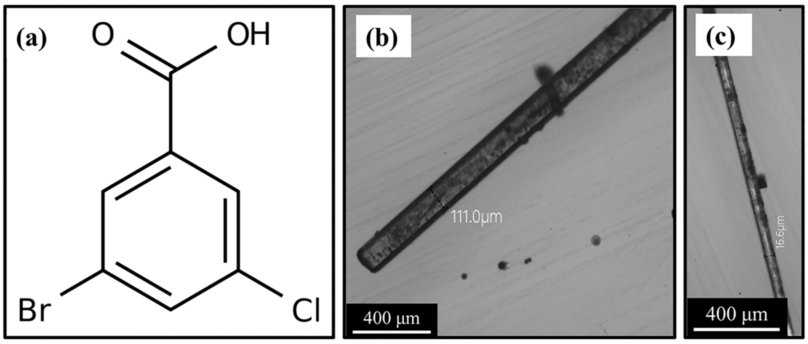

Figure 1. (a) Molecular structure of 3-bromo-5-chlorobenzoic acid (3B5CBA); optical micrograph of needle-shaped crystal of 3B5CBA with (b) wider crystallographic face, width ≈ 111.0 μm, and (c) narrower crystallographic face, thickness ≈ 16.6 μm.

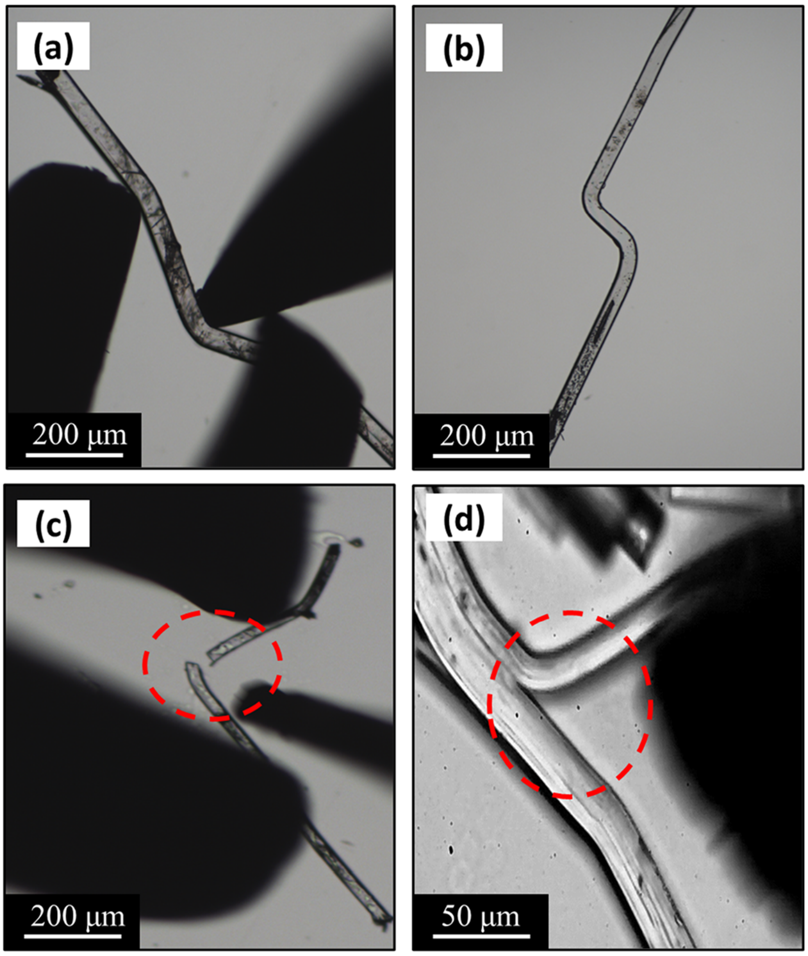

Figure 2. Optical micrograph of a needle-shaped crystal of 3B5CBA. (a) Bending of 3B5CBA crystal upon application of force on the wider crystal face, (b) plastic crystal after the removal of the applied force, (c) crystal broken into two parts with continued force on wider crystal face, marked by a red dotted circle, and (d) delamination behavior observed during bending, marked by a red dotted circle.

refined using part 1 and part 2 instructions. The site occupancy factors (SOFs) were 0.65 and 0.35 for the Br atom, and 0.88 and 0.12 for the Cl atom for part 1 and part 2, respectively, and then they were fixed at those values. Details of the data collection strategy, refinement parameters, and crystallographic data are provided in Table S2. The crystal packing diagram was generated using Mercury software (version 3.8).[54]

2.4. Sublimation Behavior of 3B5CBA Crystals. A confined microchamber was designed to observe the melting of the 3B5CBA crystals. In a confined microchamber, a circular space of a diameter of 8−10 mm was created to keep the crystal between two glass coverslips by using solid adhesive with a 1−2 mm gap between them. This confined microchamber was placed on a heating block to see the thermal response of the 3B5CBA crystals. To observe thermal response, three combinations were tried: (i) blank run, where there was no crystal sample placed in the confined microchamber and focused on the bottom surface of the top glass coverslip to ensure that observed changes in subsequent experiments were due solely to crystals. A single crystal was placed in a confined microchamber and (ii) focused on the top surface of the bottom glass coverslip to observe crystal melting and (iii) focused on the bottom surface of the top glass coverslip to observe the condensation and recrystallization of desublimated 3B5CBA crystals.

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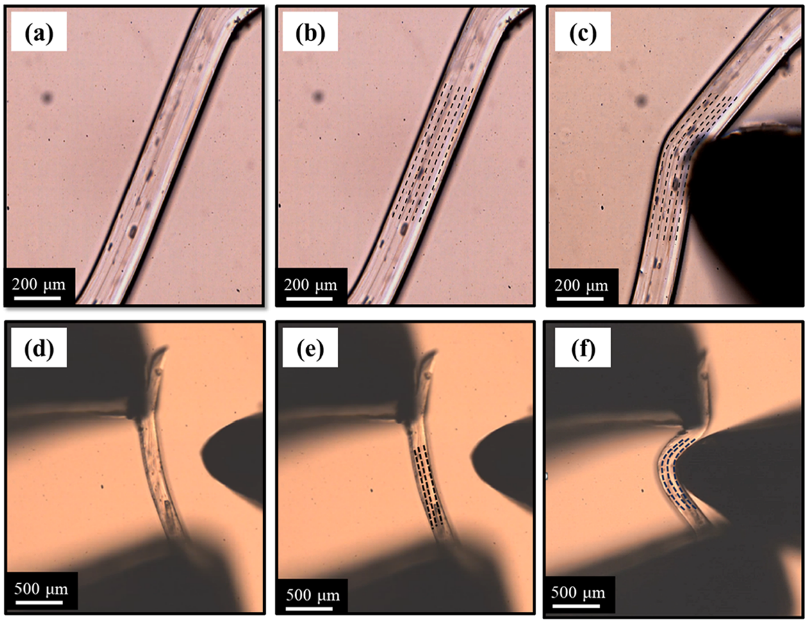

Figure 3. Optical micrographs showing the movement of striations (highlighted by black dotted lines) during mechanical deformation of 3B5CBA crystals: (a−c) Deformation of thin plastic crystals under needle-induced force and (d−f) deformation of thick plastic crystals under three-point bending. Striation shifts indicate internal structural reorganization during bending.

2.5. Energy Framework Analysis and Interaction Energy Calculations. A quantitative analysis of intermolecular interactions in needle-shaped single crystals of 3B5CBA was conducted by evaluating the total interaction energy and generating energy framework diagrams using a Crystal Explorer 21.5.[55] The calculations were performed based on the B3LYP/6−31G(d,p) molecular wave function. The energy framework diagram was constructed by considering the crystal symmetry and total interaction energy components, which included electrostatic, polarization, dispersion, and exchange-repulsion interaction components scaled by 1.057, 0.740, 0.871, and 0.618, respectively. Pairwise interaction energies between molecules were calculated within a 3.8 Å radius, clustering around the selected molecule. For clarity, interactions with energies below 5 kJ/mol were omitted. The cylinder thickness in total energy frameworks, the coulomb energy framework, and the dispersion energy framework were set to 30 kJ/mol, which is directly proportional to the intermolecular interaction energy. Crystal Explorer 21.5[55] was also used to determine the value of globularity and asphericity, for which the Hirshfeld surface was generated for the 3B5CBA, and the value of globularity and asphericity was directly obtained from the calculated surface properties provided by the software.

2.6. Nanoindentation Test. The mechanical properties, such as hardness (H) and elastic modulus (E) or reduced elastic modulus (E_r), also referred to as an effective modulus,[56] of the plastically bendable crystals at a small length scale, were quantified by a nanoindentation test. A nanoindentation test was performed on a single crystal of 3B5CBA using TI−950 Tribo Indentor (Hyistron Inc., Minneapolis, MN) with a three-sided Berkovich pyramidal tip. The tip radius for a typical Berkovich probe is ∼100 nm. Locations for indents were identified by using an optical microscope with a nanoindentation system. Large needle-shaped crystals were selected for nanoindentation, as submicrometer-sized crystals were difficult to handle and manually mount for testing. Selected crystal sample faces were cleaned in paraffin oil to create an appropriate smooth surface for nanoindentation. Before the indentation experiment, face indexing of selected crystals was performed with SCXRD to know the crystal's shape with the accurate name of the faces in terms of Miller indices (*hkl*). The selected face-indexed crystal was mounted carefully onto a substrate (metallic puck) using a very thin layer of adhesive (cyanoacrylate for molecular crystals), and there was no air gap between the crystal and the substrate. Since plasticity of the crystal was assessed using a three-point bending test, all nanoindentation experiments were performed at room temperature. Each crystal was indented at least three times on the plastically bendable (001) face with a maximum load of 1500 μN.

3. RESULTS AND DISCUSSION

While plasticity is a key property of plastic crystals, it is not exclusive to the rotator or plastic crystalline phase; it can even occur in those crystals which have no dynamic disorder.[2,9,12] There are many ordered organic crystals that also show anisotropic plastic deformation.[18,30,57] Generally, the reported

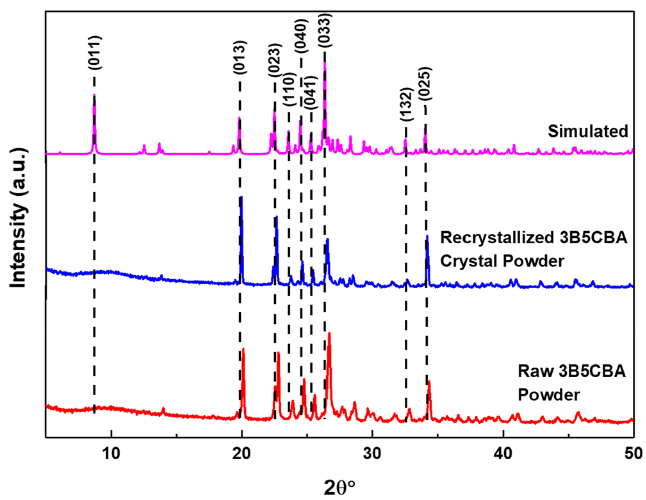

Figure 4. Comparison of powder X-ray diffraction (PXRD) patterns for 3B5CBA: simulated pattern from single-crystal X-ray diffraction (top), experimental PXRD of recrystallized crystal powder by fast evaporation (middle), and raw powder (bottom). Diffused halo pattern, slight shift in Bragg's reflection, and variation of their peak in raw 3B5CBA powder and recrystallized 3B5CBA crystal powder from simulated pattern could likely be due to molecular disorder.

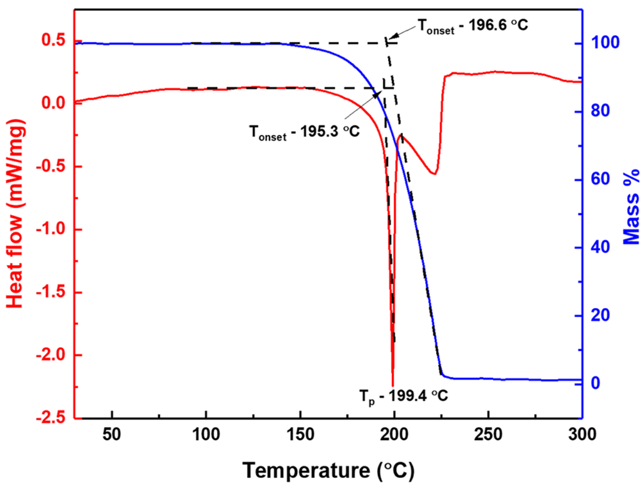

Figure 5. Thermal analysis of 3B5CBA crystals. DSC (red curve) and TGA (blue curve) thermograms were recorded at a heating rate of 5 K/min. The DSC curve showed a distinct endothermic peak with an onset temperature corresponding to the melting at 195.3 °C, while the TGA curve indicates the onset of weight loss starting around 196.6 °C.

plastic deformation crystals have a lower energy plane called a slip plane.[20,30] These planes are generated due to weak interaction in at least one crystallographic plane, allowing the necessary molecular movement for easy slippage of the column/2D molecular sheet in plastically bendable crystals under applied stress. The molecule chosen in this work, 3B5CBA, is capable of forming a carboxylic homosynthon dimer, a halogen bond, and $\pi-\pi$ stacking, as shown later in Section 3.4. Needle-shaped crystals of 3B5CBA were obtained using a fast evaporation technique (Figure S1). These crystals were found to have two major crystallographic faces: a wider crystallographic face and a narrower crystallographic face (Figure 1). Under an optical microscope, it was found that some crystals were pristine and others were opaque, whereas some crystals had voids (Figure S2). These voids were created within crystals because of the fast evaporation of DCM solvent during crystallization. The thickness, width, and length of these crystals varied in the range of 30−110 μm, 40−350 μm, and 1500−2500 μm, respectively (Figure S3).

3.1. Three-Point Bending Test. The plastic deformation behavior of 3B5CBA single needle-shaped crystals was assessed using a three-point bending test under an optical microscope. In this method, a single crystal was held by tweezers and mechanically deformed by using a fine needle. Approximately 100 crystals were selected for the bendability test, and these were divided into categories of thick and thin crystals based on their dimensions. Crystals of both types were found to deform plastically. Upon application of force on a major crystallographic face, thick crystals displayed remarkable plasticity, bending into a U or V shape without delamination (SI Video 1 and Figure 2a,b), but exhibited brittle behavior when force was applied to a narrow crystallographic face[17] (SI Video 2). Thin

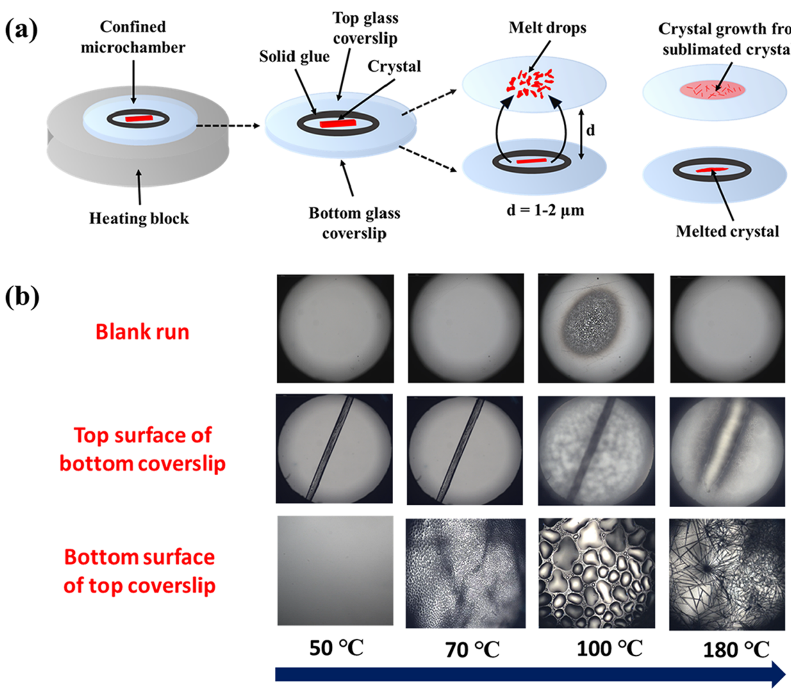

Figure 6. (a) Schematic representation of the thermal behavior of the 3B5CBA crystal analyzed using a hot-stage microscope. A confined microchamber was constructed by sealing two glass coverslips with solid adhesive, maintaining an interspace of 1−2 mm. A crystal (red) was placed within this confined microchamber. This confined microchamber was placed on the heating block. Upon heating, sublimated vapor condensed on the bottom surface of the top cooler glass coverslip. This condensed phase subsequently crystallized into needle-shaped crystals, while the original crystal on the lower glass coverslip melted as the temperature continued to rise. (b) Snapshots showing the thermal behavior of 3B5CBA crystals in a confined microchamber. A blank run was conducted without a crystal with focus on the bottom surface of the top coverslip. A crystal was placed in a confined microchamber and focused on the top and bottom coverslips; a combination of both top and bottom surface focused analysis showed that sublimation of crystals started at approximately 70 °C, followed by condensation and subsequently recrystallization of sublimated 3B5CBA into needle-shaped crystals on the bottom of the top glass coverslip. Complete melting of the original crystal on the bottom glass coverslip occurred at 180 °C.

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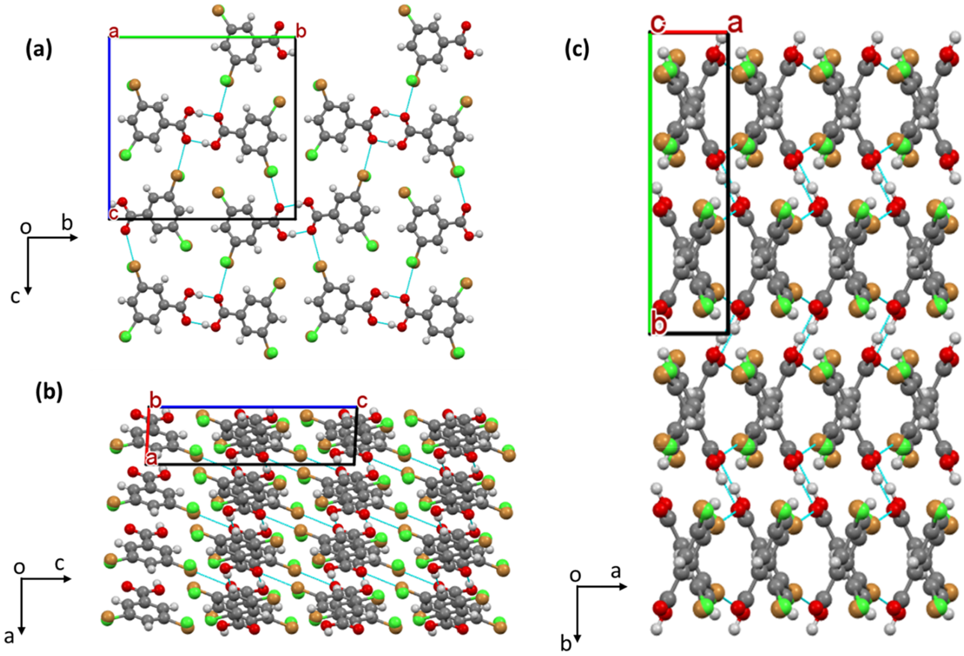

Figure 7. Molecular arrangement of 3B5CBA crystals viewed along (a) *a*-axis, (b) *b*-axis, and (c) *c*-axis.

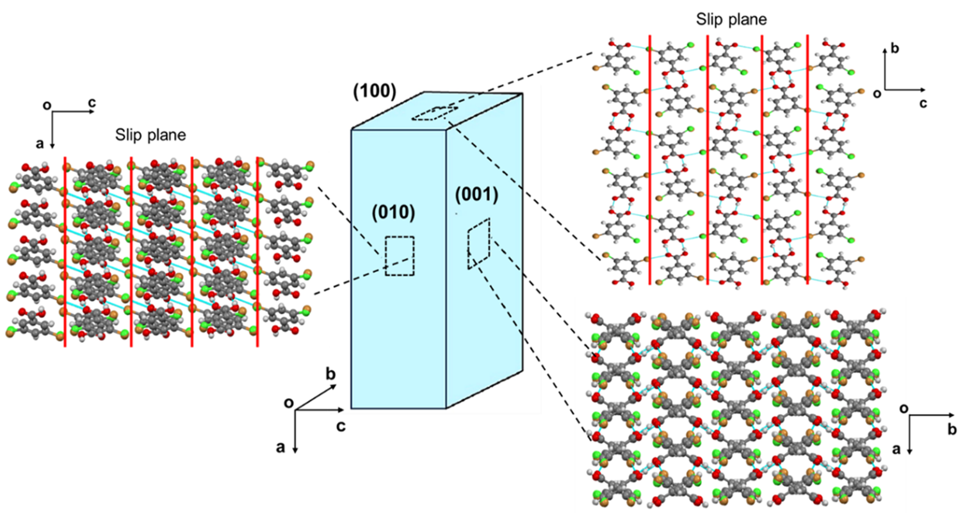

Figure 8. Crystal packing arrangement of 3B5CBA viewed on (100), (010), and (001) faces. Red lines represent the slip plane between 2D sheets.

crystals were easily deformed with minimal force and frequently exhibited a delamination behavior (SI Videos 3 and 4 and Figure 2d). Such plastic deformation could be due to the sliding of a slip plane, generally along the needle direction, leading to slightly disordered domains and delamination of layers.[58] The deformation behavior of both thin and thick crystals was recorded using a high-speed camera during the application of force. Initially, fine striations were observed on the surfaces of unbent crystals. As force was gradually applied, noticeable movement of these striations (parallel lines) was observed in thin crystals (Figure 3a−c and SI Video 3) and thick crystals (Figure 3d−f and SI Video 5). Upon the removal of the force, the striations were stabilized and remained in their final positions, providing direct insight into the mechanical response and striation dynamics of plastically deforming crystals. These plastic crystals were broken into two parts when the applied force exceeded a certain limit (Figure 2c and SI Video 6), indicating that plasticity depends upon the direction and amount of force applied. Furthermore, crystals were found to bend from multiple locations on application of force, suggesting a high degree of uniformity in mechanical behavior throughout the crystal (SI Video 7). SEM and AFM analyses showed that the application of force by the three-point bending test introduces discernible surface irregularities on the (010) face of the 3B5CBA crystal (Figures S4 and S5).

3.2. Powder X-ray Diffraction Analysis. Powder X-ray diffraction (PXRD) patterns of 3B5CBA were collected for the raw powder and crystal powder (obtained after solid-state grinding of recrystallized 3B5CBA) and compared with the simulated pattern generated from single-crystal X-ray diffraction (SCXRD) data (Figure 4). The PXRD pattern of the raw 3B5CBA and recrystallized crystal powder shows a good agreement with the simulated pattern, confirming phase purity and successful recrystallization of the same compound. Nevertheless, one additional reflection observed in the

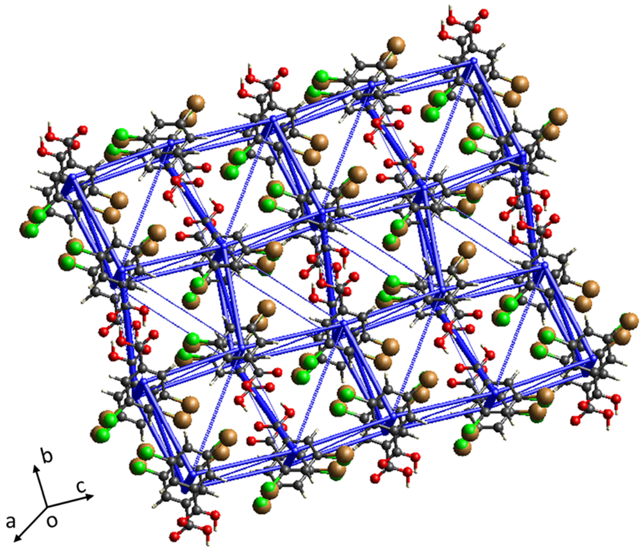

Figure 9. 3D topologies of the total interaction energy framework diagram of the 3B5CBA crystal structure. The tube size and threshold value were set at 30 kJ/mol.

simulated pattern at (011) was absent, and a diffused halo was present in both experimental PXRD patterns of raw powder and recrystallized crystal powder of 3B5CBA. Additionally, the PXRD pattern of raw 3B5CBA powder and recrystallized 3B5CBA crystal powder exhibited a slight shift in the positions of Bragg's reflections and variation of their peak intensities relative to the simulated pattern. This could likely be due to the incorporation of molecular disorder in both raw and recrystallized 3B5CBA powder samples.[59,60] Further, in the PXRD pattern of raw powder and the powder of recrystallized 3B5CBA crystals, a peak corresponding to $2\theta \approx 13°$ (present in the simulated XRD pattern) appears to be absent. This could be due to the random orientation of small crystallites in the crushed samples of 3B5CBA, which often leads to peak broadening and loss of low-intensity reflections within background noise during measurement.[61] PXRD patterns of crystals grown in various solvents were also recorded and found to be nearly identical (Figure S6), indicating that all samples had the same molecular packing arrangement irrespective of the solvent used during crystallization. This suggests that the observed plastic bendability of the crystals was governed by the intrinsic molecular arrangement rather than by the choice of solvent during growth.

3.3. Thermal Behavior of 3B5CBA Crystals. The thermal behavior of 3B5CBA needle-shaped crystals was systematically evaluated using TG-DSC and HSM analysis. As observed from Figure 5, 3B5CBA crystals began to undergo weight loss at approximately 140 °C, attributed to sublimation, with ~25% of the total weight lost before melting. Melting corresponds to an endothermic peak with an onset at 195.3 °C and a peak temperature of 199.4 °C. Correspondingly, the TGA curve indicated a sharp weight loss beginning near the melting point.

To visualize the thermal response of 3B5CBA crystals, an HSM experiment was further conducted. Upon heating, crystal size gradually decreased as the crystal transitioned from solid to vapor phase when the temperature reached 70 °C, indicating sublimation (SI Video 8 and Figure S7a,b). Eventually, needle-shaped crystals were observed to have grown on the glass coverslip, attached to the viewing window by a desublimation process (SI Video 9 and Figure S7c,d). These findings demonstrate that sublimation occurred at a significantly lower temperature than the melting point of 195.3 °C (as determined by DSC), due to its plastic nature. To observe the melting of 3B5CBA crystals, an experiment was further conducted in a confined microchamber, as mentioned in Section 2.4 and depicted as a schematic in Figure 6a. In Figure 6b, at approximately 100 °C in the blank run, moisture was released from the solid adhesive and observed on the top glass coverslip. With the continued increase in temperature, condensed moisture evaporated and a clean surface appeared. A crystal of 3B5CBA was placed inside the confined microchamber to observe melting. As the temperature increased, there were noticeable changes that appeared on the outer surface of crystals at 68−70 °C because of sublimation. A continued increase in temperature results in the complete melting of single crystals observed around 180 °C and the recrystallization of needle-shaped 3B5CBA crystals on the bottom surface of the top glass coverslip, following the vapor-to-melt-to-crystal growth mechanism because of the

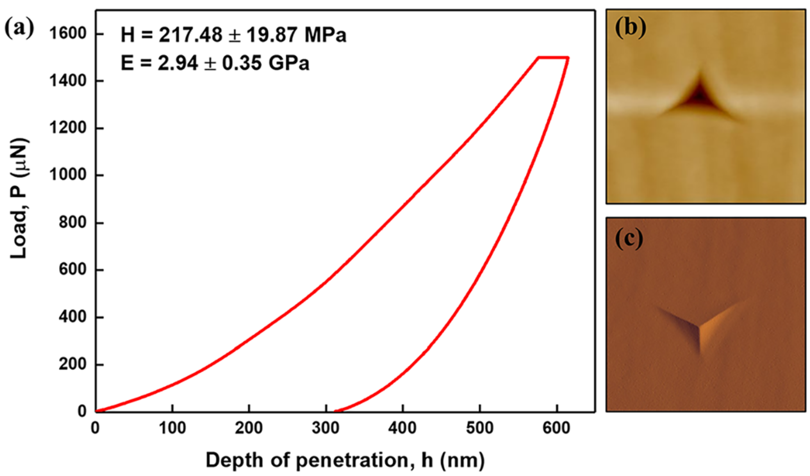

Figure 10. (a) Load−depth of penetration (P−h) curve obtained from the nanoindentation test with a maximum load of 1500 μN on the (001) face of the 3B5CBA single crystal. (b) Scanning probe images (topographical). (c) Scanning probe image of Berkovich indents on the (001) face of the same single crystal in a scan size of 2 μm.

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Table 1. Comparison of Elastic Modulus (E) and Hardness (H) Values of Brittle, Elastic, and Plastic Crystals of Organic Molecular Crystals Reported in Literature

sr. no.	molecule name	mechanical behavior	E (GPa)	H (MPa)	plane used for nanoindentation	refs
1.	*N*,*N*′-dipropyl-1,4,5,8-naphthalenetetracarboxylic diimide (1nPr)	brittle	8.49 ± 0.09	399 ± 7	(001)	27
			10.18 ± 0.24	427 ± 22	(010)	
2.	difluoroboron avobenzone (BF_2AVB)	brittle-form I	10.6 ± 0.35	275 ± 12	(011)	64
			7.4 ± 0.31	340 ± 7	(001)	
			13.2 ± 0.16	410 ± 11	($\bar{1}$20)	
3.	1,2,4,5-tetrabromobenzene	brittle-β form	6.414 ± 0.124	672 ± 9	(1$\bar{1}$0)	65
4.	difluoro(3-(hydroxy)-3-(4-methoxyphenyl)-1-phenylprop-2-en-1-onato)borate [BF_2dbm(OMe)]	brittle	8.620 ± 0.176	255.79 ± 8.48	(001)	66
5.	saccharin	brittle	13.94 ± 0.07	530 ± 10	(100)	67
6.	sulfathiazole form 2	brittle	21.88 ± 0.22	900 ± 30	(100)	
7.	L-alanine	brittle	25.1 ± 0.72	990 ± 50	(001)	
8.	3-((4-chlorophenyl)imino)-indolin-2-one	brittle-form I	8.09 ± 0.2	330 ± 90	(010)	68
9.	ethyl[(4*Z*)-2-methyl-5-oxo-4-(2,3,4-trimethoxybenzylidene)-4,5-dihydro-1*H*-imidazol-1-yl]acetate	brittle	8.0 ± 0.51	190 ± 41	(010)	69
10.	bazedoxifene acetate	brittle-form A	8.02 ± 0.75	300 ± 30	(010)	70
		brittle-form B	10.37 ± 0.22	390 ± 10	(10$\bar{1}$)	
		brittle-form D	12.60 ± 0.11	590 ± 8	(00$\bar{1}$)	
11.	carbamazepine	brittle-form III	13.87 ± 0.26	470 ± 17	(020)	71
12.	(*E*)-*N*′-(4-methylbenzylidene)picolinohydrazide (MBPH)	brittle	5.502 ± 0.082	201 ± 5	(001)	72
13.	(*E*)-2-(4-chlorobenzylidene)-2,3-dihydroinden-1-one	plastic	3.45	160	(001)	18
14.	(*E*)-2-(4-chlorobenzylidene)-5-bromo-2,3-dihydroinden-1-one	plastic	0.45	35	(001)	18
15.	(*E*)-2-(4-bromobenzylidene)-5-bromo-2,3-dihydroinden-1-one	plastic	8.98	204	(001)	18
16.	(*E*)-2-(3,4-dichlorobenzylidene)-2,3-dihydroinden-1-one	plastic	9.56	382	(001)	18
17.	*N*,*N*′-dimethyl-1,4,5,8-naphthalenetetracarboxylic diimide(1Me)	plastic	1.82 ± 0.10	223 ± 7	(001)	27
18.	coumarin	plastic-form II	0.846 ± 0.022	39 ± 22	(010)	32
19.	flufenamic acid	plastic-form III	7.90 ± 0.18	110 ± 2	(100)	46
20.	difluoroboron avobenzone (BF_2AVB)	shearing-form II	5.6 ± 0.46	206 ± 13	(001)	64
21.	(1,3-bis(4-*t*-butylphenyl)-3-(hydroxy)prop-2-en-1-onato)(difluoro)borate [BF_2dbm(^{t}Bu)$_2$]	plastic	0.369 ± 0.008	92.45 ± 4.04	(001)	66
22.	difluoro(3-(hydroxy)-1,3-bis(4-methoxyphenyl)prop-2-en-1-onato)borate [BF_2dbm(OMe)$_2$]	shearing	10.864 ± 0.249	264.93 ± 10.98	(010)	
23	3-((4-chlorophenyl)imino)-indolin-2-one	plastic-form II	3.54 ± 0.05	130 ± 15	(100)	68
24.	ethyl[(4*Z*)-2-methyl-5-oxo-4-(2,3,4-trimethoxybenzylidene)-4,5-dihydro-1*H*-imidazol-1-yl]acetate	plastic	5.3 ± 0.22	170 ± 5	(001)	69
25	isonicotinamide	plastic-form I	12.110 ± 0.515	278 ± 21	(100)	76
		plastic-form II	20.262 ± 0.660	498 ± 27	(100)	
		plastic-form V	15.147 ± 2.985	1026 ± 337	(100)	
26.	4-trifluoromethyl phenylisothiocynate	plastic	9.56 ± 1.1	650 ± 14	(001)	77
			5.72 ± 0.8	410 ± 15	(101)	
27.	3,6-dibromo-9-ethyl-9*H*-carbazole	plastic	6.86 ± 0.42	160 ± 13	(001)	78
28.	2-((*E*)-(6-methylpyridin-2-ylimino)methyl)4-chlorophenol	plastic	8.4 ± 1	317 ± 33	(100)	79
29.	2-((*E*)-(6-methylpyridin-2-ylimino)methyl)4-bromophenol	plastic	4.54 ± 0.45	204 ± 167	(100)	79
30.	2-((*E*)-(6-bromolpyridin-2-ylimino)methyl)4-bromophenol	plastic	7.05 ± 0.25	214 ± 10	(010)	79
31.	9-anthraldehyde	elastic	6.74 ± 0.15	140 ± 7	(001)	23
			6.31 ± 0.08	130 ± 5	(010)	
32.	*N*,*N*′-diethyl-1,4,5,8-naphthalenetetracarboxylic diimide(1Et)	elastic	16.10 ± 0.3	502 ± 38	(001)	27
33.	coumarin	elastic-form I	2.501 ± 0.172	79 ± 11	(100)	32
34	3-((4-chlorophenyl)imino)-indolin-2-one	elastic-form III	11.5 ± 1.2	380 ± 50	(001)	68
35.	(*E*)-*N*′-benzylidenepicolinohydrazide (BPH)	elastic	4.832 ± 0.373	262 ± 11	(002)	72
36.	1,3-diamino-2,4,5,6-tetrabromobenzene	elastic	12.3 ± 0.3	480 ± 10	(100)	74
37.	4,4'-diazenediylbis{*N*,*N*-bis[(pyridine-2-yl)methyl]benzamide} tetrahydrate	elastic	11.64 ± 0.46	352 ± 14	(100)	75

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Table 1. continued

sr. no.	molecule name	mechanical behavior	E (GPa)	H (MPa)	plane used for nanoindentation	refs
38.	3-bromo-5-chlorobenzoic acid	plastic	2.94 ± 0.35	217.48 ± 19.87	(001)	this work

temperature gradient within the confined microchamber,[62] where the top glass coverslip remained cooler than the bottom one. To verify that sublimation occurred without decomposition, FTIR and Raman analyses were performed for both the 3B5CBA crystal powder grown by fast evaporation at room temperature and desublimation in a Linkam chamber. As shown in Figure S8, FTIR and Raman spectra confirmed that only sublimation occurred without decomposition during temperature increase.

3.4. Estimation of Crystal Structure of 3B5CBA Crystals. A single-crystal structural analysis was conducted to carry out face indexing and understand molecular packing in the 3B5CBA crystals. The crystal structure was found to be monoclinic with the $P2_1/c$ space group. The unit cell contained four molecules, with one in the asymmetric unit (Table S2), with the molecule exhibiting occupancy disorder at the halogen site. Face indexing of a single crystal was performed to identify the crystal faces, with a wider crystallographic face indexed as (001) and a narrower face as (010) (Figure S9). In the crystal structure (Figure 7), 3B5CBA molecules formed a homodimer connected through carboxylic acid homosynthon along the *b*-axis via strong intermolecular O−H···O hydrogen bonds (D, d, θ: 2.644 Å, 1.722 Å, 178.25°, respectively). These dimers were stacked through $\pi-\pi$ stacking along the *a*-axis, with a centroid-to-centroid distance of 3.912 Å between two aromatic rings. The combination of aromatic interactions and strong hydrogen bonding formed a stable, 2D sheet-like structure. These 2D sheets interacted with the adjacent sheet via weak bonds C−Br···O (3.055 Å, 175.354°) interactions along the (001) plane. These weak interactions resulted in the formation of a slip plane parallel to the (001) face indicated by a red line (Figure 8). The slip planes facilitated easy sliding and separation of 2D sheets when force was applied on the (001) face on the straight crystal. This phenomenon was responsible for bending and delamination in needle-shaped single crystals of 3B5CBA.

3.5. Globularity, Asphericity, and Pairwise Interaction Energy Calculations. Plastic crystals are often composed of molecules with nearly spherical shapes, making molecular sphericity a useful structural criterion for identifying potential plastic crystal formers.[3] The globularity (G) quantifies how closely a molecule's shape approximates a sphere by comparing its Hirshfeld surface area to that of a sphere with the same volume. Asphericity (Ω), on the other hand, measures molecular anisotropy based on the principal moments of inertia, with a value of zero indicating a perfectly spherical object.[6] For 3B5CBA, the calculated values are $G = 0.808$ and $\Omega = 0.108$, using a method mentioned in Section 2.5, which indicate high globularity and low asphericity, generally associated with plastic crystal-forming compounds.

Since slip planes are the lower energy plane, the pairwise energy of interaction was estimated to identify the slip plane. The main reason for calculating the pairwise interaction energy was to correlate it to the observed bending behavior of 3B5CBA crystals. As such, the magnitude of the interaction energy represents the strength, where a more negative value represents a stronger interaction. In Figure 9, the arrangement of molecules was shown along with their interaction energy. The aggregate of total energies was considered separately for (i) carboxylic acid homosynthon, (ii) $\pi-\pi$ stacking, and (iii) halogen bonding. Carboxylic homosynthon and $\pi-\pi$ stacking together formed a 2D plane in [100] and [010], with a strong interaction energy of −299.9 kJ/mol. These 2D planes interacted with each other through C−Br···O interaction along [001] with an interaction energy of −79.6 kJ/mol (Figures S10 and S11, and Table S3). Hence, the plane formed by the C−Br···O interaction is likely to be a facile slip plane along which bending or delamination of crystals occurs.

3.6. Analysis of Mechanical Properties through Nanoindentation. Figure 10 presents a representative load−displacement ($P-h$) curve, along with scanning probe microscope (SPM) images for the plastically bendable (001) face of 3B5CBA crystals under the peak load of 1500 μN. The loading segment exhibited a smooth curve with no evidence of pop-ins, and the corresponding SPM images showed no sign of cracking or fracture, indicating uniform plastic deformation.[23] The calculated values of stiffness (S), hardness (H), Young's modulus (E), and reduced elastic modulus (E_r) were 9.98 ± 0.72 μN/nm, 217.48 ± 19.87 MPa, 2.94 ± 0.35 GPa, and E_r = 3.22 ± 0.38 GPa, respectively (see SI section S1 for calculation). The minimal difference between E_r and E values indicates that the indenter's compliance was negligible during measurement.[56] Values of E and H indicate crystals of 3B5CBA are supercompliant and soft molecular crystals (Table S4).[63] Further, these values of E and H for 3B5CBA crystals compare well with the reported values in the literature for plastic crystals (Table 1).[64−79]

4. CONCLUSIONS

In this work, the plastic behavior of 3B5CBA has been investigated. Long, high-quality, and needle-shaped crystals of 3B5CBA were grown by a fast evaporation technique in various organic solvents such as DCM and toluene. These crystals were observed to bend and delaminate during the three-point bending test when force was applied parallel to the wider crystallographic face of (001). PXRD pattern confirmed that recrystallized crystals retained identical packing across crystals grown from various solvents, suggesting that bendability arises from packing rather than solvent choice. Thermal analyses using TGA, DSC, and HSM showed sublimation of crystals beginning around 70 °C in a confined microchamber kept in HSM and 30% of the total weight loss before melting at 195.3 °C in TG-DSC. HSM studies revealed sublimation−driven recrystallization within a confined microchamber, influenced by vacuum or reduced pressure. SCXRD analysis and interaction energy framework diagram demonstrated the presence of strong interaction in a 2D sheet of molecules parallel to the (001) plane with an interaction energy of −299.9 kJ/mol. These 2D sheets interacted with each other with C−Br···O interaction in [001] with an interaction energy of −79.6 kJ/mol, which is low compared with other interactions present. Hence, the plane formed by the C−Br···O interaction was likely to be a facile slip plane. The nanoindentation test on the (001) face revealed homogeneous plastic deformation with no pop-in or fracture; the measured

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hardness and Young's modulus were 217.48 ± 19.87 MPa and 2.94 ± 0.35 GPa, respectively.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.cgd.5c01256.

Preparation of single crystals of 3B5CBA, optical micrographs of needle-shaped single crystals of 3B5CBA, surface morphology analysis by SEM and AFM, PXRD analysis, sublimation of 3B5CBA crystals, Raman and FTIR analysis, crystallographic parameters, face indexing, energy framework analysis, nanoindentation (PDF)

Plastic bending of a thick crystal (Video S1) (MP4)

Brittle nature of single crystals (Video S2) (MP4)

Plastic bending and movement of striations in a thin single crystal (Video S3) (MP4)

Delamination of a thin plastic crystal (Video S4) (MP4)

Movement of striations in a thick crystal (Video S5) (MP4)

Continued application of force broke the crystal into two parts (Video S6) (MP4)

Plastic bending of a crystal on the application of force stress on the wider face (001) at different locations (Video S7) (MP4)

Sublimation of single crystals (Video S8) (MP4)

Desublimation of needle-shaped crystals on the top glass coverslip in a Linkam chamber (Video S9) (AVI)

Accession Codes

Deposition Number 2488286 contains the supporting crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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